



Top 13 CAT Routes and Networks Questions With Video Solutions

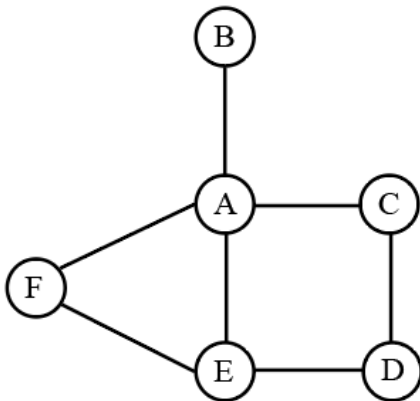
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Questions

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 1

Which one among the following stations is visited the largest number of times?

- A Station C
- B Station E
- C Station D
- D Station F

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known:

1. None of the streets has more than one team traveling along it in any direction at any point in time.
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4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

Question 2

How many times do the teams pass through Station B in a day?

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can pass via Station A at any point on their journeys. The following facts are known:

- Team 1 has more than one team travelling along it in any direction at 10:00 hrs.
- Team 2 is in stations E and D respectively at 10:00 hrs.
- Team 3 is in station E at 10:30 hrs.
- Team 4 is in stations B and E respectively at 11:30 hrs.
- Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
- Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

Question 3

Which team patrols the street connecting Stations D and E at 10:15 hrs?

- A Team 4
- B Team 1
- C Team 2
- D Team 3

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can pass via Station A at any point on their journeys. The following facts are known:

- Team 1 has more than one team travelling along it in any direction at 10:00 hrs.
- Team 2 is in stations E and D respectively at 10:00 hrs.
- Team 3 is in station E at 10:30 hrs.
- Team 4 is in stations B and E respectively at 11:30 hrs.
- Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
- Team 4 never passes through Stations B, D or F.

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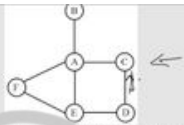
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Question 4

How many times does Team 4 pass through Station E in a day?



	9	9:50	10	10:30	11	11:30	12
Team 1	A		E				A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A



The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can only move in the direction of the arrows on their journeys.

Additional facts are known:

- There is no team traveling along it in any direction at 10:00 hrs.
- Teams 1 and 2 are the only teams that pass through Stations E and D respectively at 10:30 hrs.
- Teams 3 and 4 are the only teams that pass through Stations B and E respectively at 11:30 hrs.
- Teams 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
- Team 4 never passes through Stations B, D or F.

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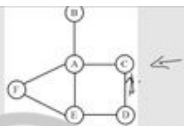
Question 5

How many teams pass through Station C in a day?

- A 4
- B 3
- C 1
- D 2



	9	9:50	10	10:30	11	11:30	12
Team 1	A		E				A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A



The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can only move in the direction of the arrows on their journeys.

Additional facts are known:

- There is no team traveling along it in any direction at 10:00 hrs.
- Teams 1 and 2 are the only teams that pass through Stations E and D respectively at 10:30 hrs.
- Teams 3 and 4 are the only teams that pass through Stations B and E respectively at 11:30 hrs.
- Teams 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
- Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

Instructions

Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from M to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute.

The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.

Question 6

How many cars would be asked to take the route A-N-B, that is Akala-Nanur-Bakala route, by the police department?

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Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from N to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute. The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.



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Question 7

If all the cars follow the police order, what is the difference in travel time (in minutes) between a car which takes the route A-N-B and a car that takes the route A-M-B?

- A 1
- B 0.1
- C 0.2
- D 0.9

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Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from N to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute. The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.



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Question 8

A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.

How many cars would the police department order to take the A-M-N-B route so that it is not possible for any car to reduce its travel time by not following the order while the other cars follow the order? (Assume that the police department would never order all the cars to take the same route.)

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Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from M to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute. The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.



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Question 9

A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by j minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.

If all the cars follow the police order, what is the minimum travel time (in minutes) from A to B? (Assume that the police department would never order all the cars to take the same route.)

- A 26
- B 32
- C 29.9
- D 30

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Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from M to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute. The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.



VIDEO SOLUTION

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Instructions

A new airlines company is planning to start operations in a country. The company has identified ten different cities which they plan to connect through their network to start with. The flight duration between any pair of cities will be less than one hour. To start operations, the company has to decide on a daily schedule. The underlying principle that they are working on is the following:
Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.

Question 10

If the underlying principle is to be satisfied in such a way that the journey between any two cities can be performed using only direct (non-stop) flights, then the minimum number of direct flights to be scheduled is:

- A 45
- B 90
- C 180
- D 135



10 cities $A \rightarrow B$ $M A \rightarrow B$ $B \rightarrow A$
 $E B \rightarrow A$ $A \rightarrow B$

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Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.

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Question 11

Suppose three of the ten cities are to be developed as hubs. A hub is a city which is connected with every other city by direct flights each way, both in the morning as well as in the evening. The only direct flights which will be scheduled are originating and/or terminating in one of the hubs. Then the minimum number of direct flights that need to be scheduled so that the underlying principle of the airline to serve all the ten cities is met without visiting more than one hub during one trip is:

- A 54
- B 120
- C 96
- D 60

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10 cities $A \rightarrow B$ $M A \rightarrow B$ $B \rightarrow A$
 $E B \rightarrow A$ $A \rightarrow B$

A new airlines company is planning to start operations in a country. The company has identified ten different cities which they plan to connect through their network to start with. The flight duration between any pair of cities will be less than one hour. To start operations, the company has to decide on a flight schedule. The underlying principle that they are working on is the following: Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.



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Question 12

Suppose the 10 cities are divided into 4 distinct groups G1, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that G1 consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:

1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4

Then the minimum number of direct flights that satisfies the underlying principle of the airline is:

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10 cities $A \rightarrow B$ $M A \rightarrow B$ $B \rightarrow A$
 $E B \rightarrow A$ $A \rightarrow B$

A new airlines company is planning to start operations in a country. The company has identified ten different cities which they plan to connect through their network to start with. The flight duration between any pair of cities will be less than one hour. To start operations, the company has to decide on a flight schedule. The underlying principle that they are working on is the following: Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.



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Question 13

Suppose the 10 cities are divided into 4 distinct groups G1, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that G1 consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:

1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4

However, due to operational difficulties at A, it was later decided that the only flights that would operate at A would be those to and from B. Cities in G2 would have to be assigned to G3 or to G4.

What would be the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose?

Routes $A \rightarrow B$ $M A \rightarrow B$ $B \rightarrow A$
 $E B \rightarrow A$ $A \rightarrow B$

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